Experiment No. – 1				
Date of Performance:	**15/1/2026**			
Date of Submission:	**22/1/2026**			
Execution (06)	Viva (02)	Professional Ethics (02)	Experiment Total (10)	Sign with Date

Experiment No. 1

1.1 Aim: Installation of Truffle and set up configuration

1.2 Course Outcome: Develop the application using the concept of Ethereum with smart contracts using different development tools.

1.3 Learning Objectives: Examine Truffle tool

1.4 Requirement:

- Computer/Laptop
- Windows / Linux / macOS
- Node.js & npm
- Truffle framework
- Ganache (local blockchain)
- Code editor (VS Code)
- Internet connection

1.5 Related Theory:

Blockchain application development requires a structured framework that simplifies smart contract creation, compilation, deployment, and testing. Truffle is a popular development framework designed to support the complete lifecycle of Ethereum-based decentralized applications (DApps). It is widely used in academic labs and industry due to its reliability, modular structure, and ease of use.

Truffle is developed and maintained by ConsenSys, a leading blockchain technology company. It provides an integrated environment that helps developers manage smart contracts written in Solidity and interact with Ethereum networks efficiently.

Need for Truffle
Developing smart contracts directly on a blockchain network is complex and error-prone due to:

- High deployment costs
- Irreversibility of transactions
- Difficulty in debugging and testing

Truffle addresses these issues by offering

- A local blockchain simulation for testing
- Automated compilation and deployment
- Scriptable migrations
- Built-in testing framework

Truffle Framework Architecture
The Truffle framework consists of the following major components:

1. Truffle CLI (Command Line Interface)
   Enables developers to create projects, compile contracts, deploy them to networks, and run tests using simple commands.
2. Smart Contract Compilation
   Truffle uses the Solidity compiler (solc) to convert smart contract code into bytecode and Application Binary Interface (ABI), which are necessary for blockchain deployment.
3. Migration System
   Migrations are JavaScript files that handle the deployment of smart contracts in a controlled and versioned manner. This ensures consistency across different networks.
4. Network Configuration
   The truffle-config.js file allows developers to define network settings such as:
   - Host address
   - Port number
   - Network ID
   - Gas limit and gas price
5. Testing Environment
   Truffle supports automated testing using JavaScript or Solidity, helping verify contract logic before real deployment.

Role of Node.js and npm
Truffle runs on Node.js, which provides a runtime environment for executing JavaScript outside the browser. The Node Package Manager (npm) is used to install Truffle globally and manage project dependencies.

Local Blockchain Setup
For development and testing purposes, Truffle is commonly used with a local Ethereum blockchain simulator (such as Ganache). This allows:

- Instant mining of transactions
- Multiple test accounts with virtual Ether
- Safe experimentation without real financial cost

Importance of Configuration Setup
Proper configuration ensures:

- Correct network connection
- Successful contract deployment
- Accurate testing results
- Smooth interaction between smart contracts and blockchain networks

The configuration file acts as a bridge between the Truffle framework and the target Ethereum network.

Truffle Suite:
Truffle Suite is a development environment based on Ethereum Blockchain, used to develop Distributed Applications. Truffle provides us with a developing environment in which we can make our decentralized environment very easy.

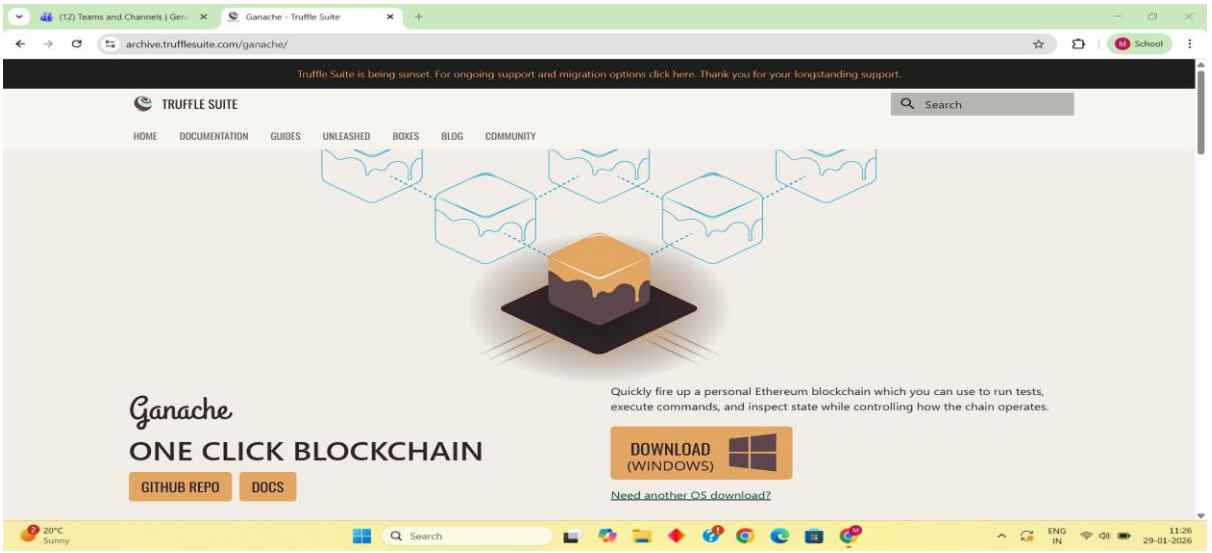
Truffle offers several different features:
Smart Contract Management: This means that Truffle helps manage the artifacts of all smart contracts utilized in your dApps. As Truffle takes care of this, you can focus on other parts of the development process. This further means that Truffle supports library linking, custom deployments, and more complex Ethereum dApps.
Automated Contract Testing: Another helpful feature of Truffle is that it supports automated contract testing. This means that you can bring your developer experience to the 21st century and construct automated tests for all your contracts. The main benefit of this is that you can shorten the development process of your smart contracts.
Scriptable Migration and Deployment: With Truffle, you can write deployment scripts that allow you to account for the fact that your dApps will change over time. This means that you can maintain your smart contracts far into the future and in the long term.
Network Management: Truffle helps you take care of your network by managing your artifacts, allowing you to focus your attention on other tasks.
Interactive Console: Truffle features an interactive console that allows you to access all Truffle commands and contracts you have built.
Ganache: Ganache is a private Ethereum blockchain environment that allows you to emulate the Ethereum blockchain so that you can interact with smart contracts in your own private blockchain. Here are some features that Ganache provides:

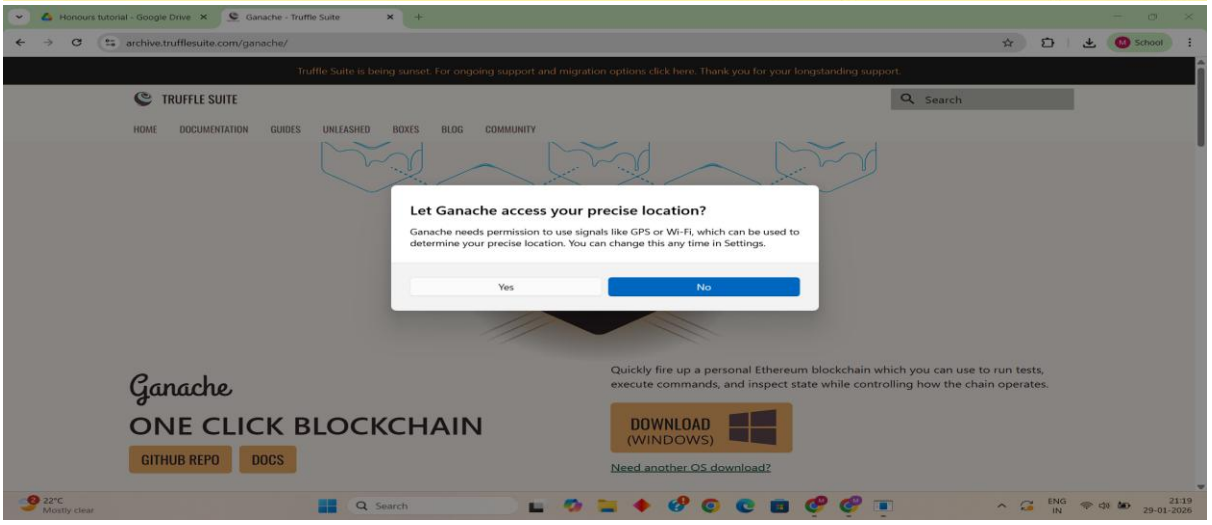
1) Displays blockchain log output
2) Provides advanced mining control
3) Built-in block explorer
4) Ethereum blockchain environment
5) Ganache has a desktop application as well as a command-line too

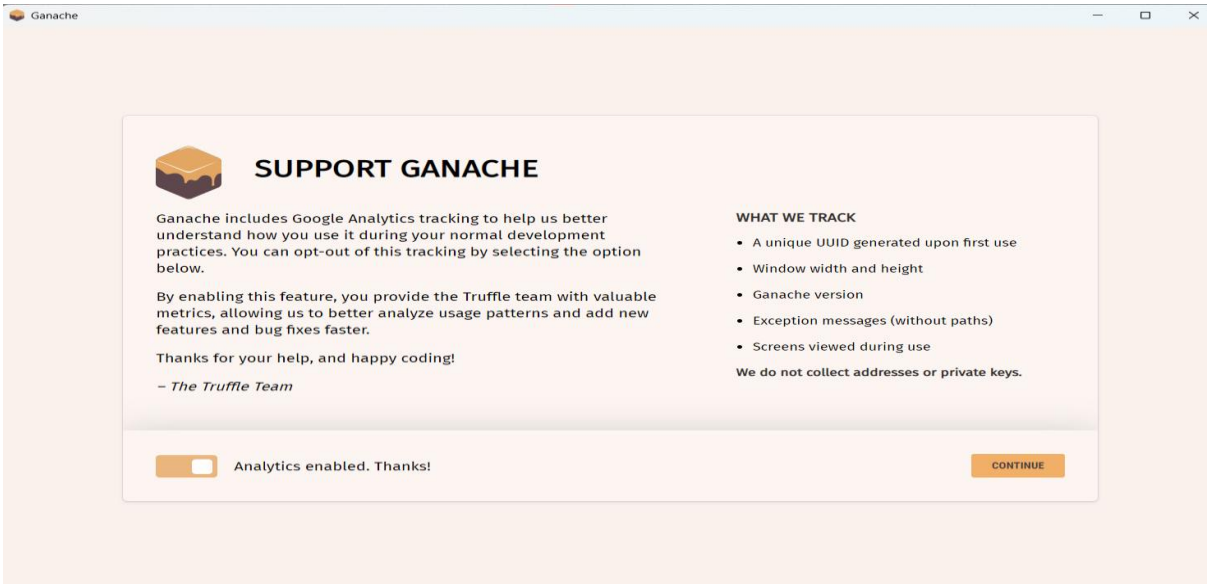
1.6 Procedure:

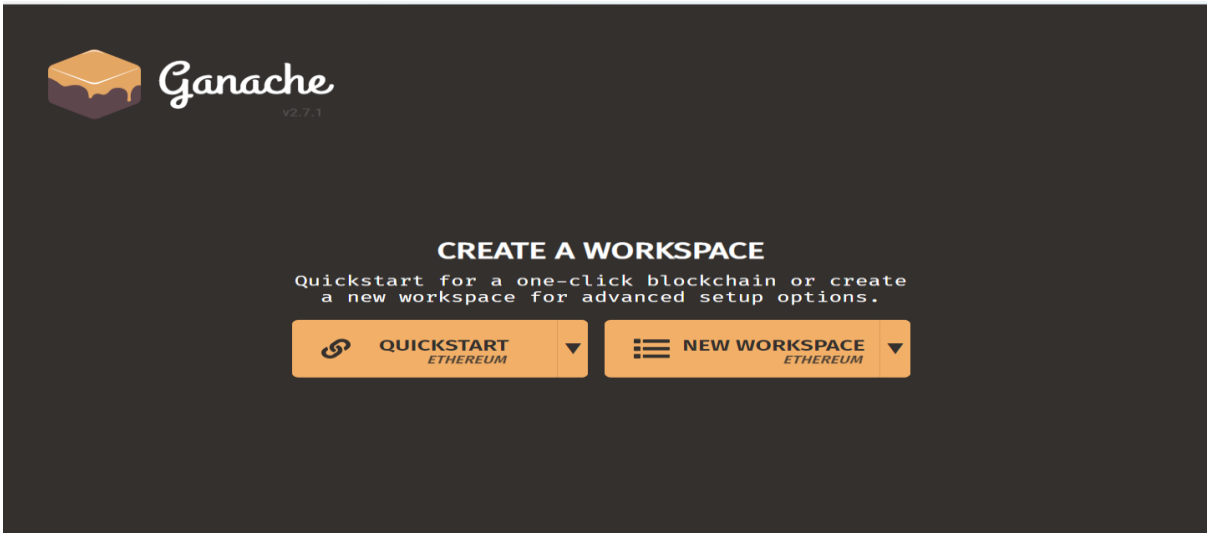
- Open a web browser and download Node.js from the official website.
- Install Node.js and verify the installation using the commands node -v and npm -v.
- Open the command prompt or terminal.
- Install Truffle globally using the command:
  npm install -g truffle
- Verify Truffle installation using:
  truffle version
- Download and install Ganache for setting up a local blockchain.
- Create a new project folder and open it in the terminal.
- Initialize a Truffle project using:
  truffle init
- Configure the truffle-config.js file to connect Truffle with Ganache network.
- Start Ganache and ensure the network settings match the configuration file.
- The Truffle environment is now successfully installed and configured.

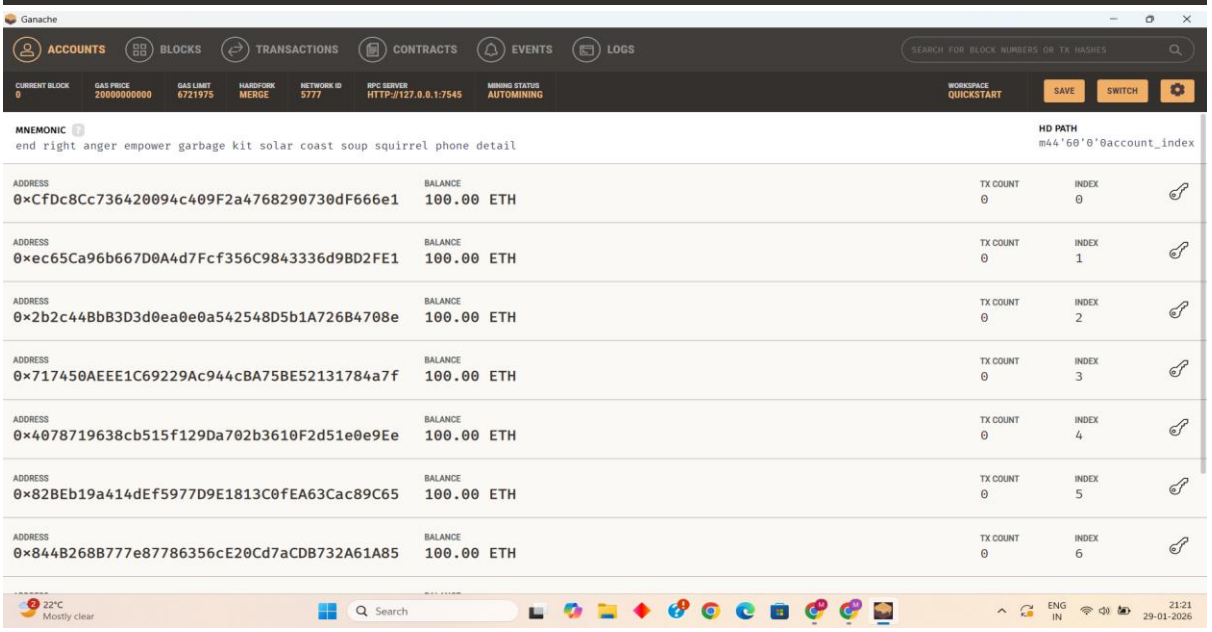
1.7 Program and Output:

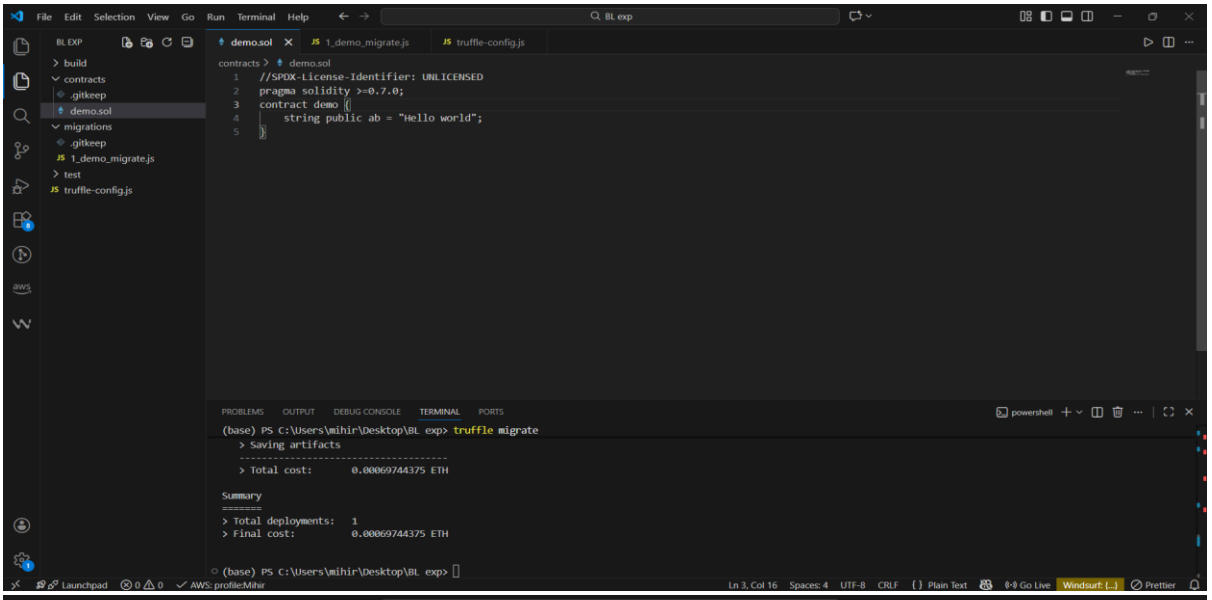


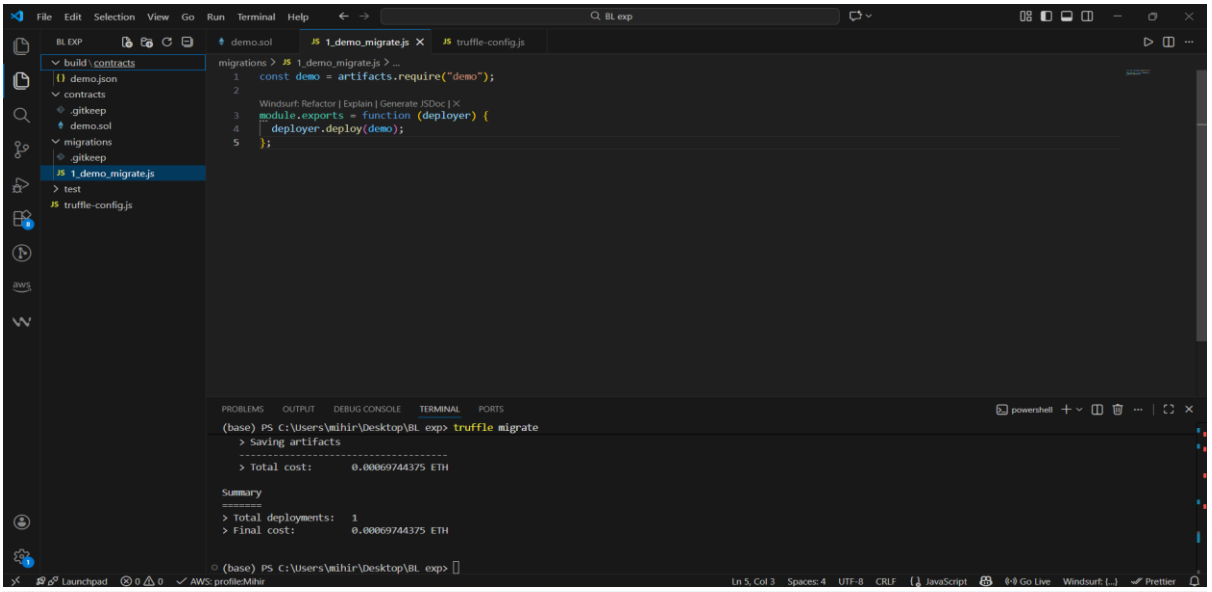


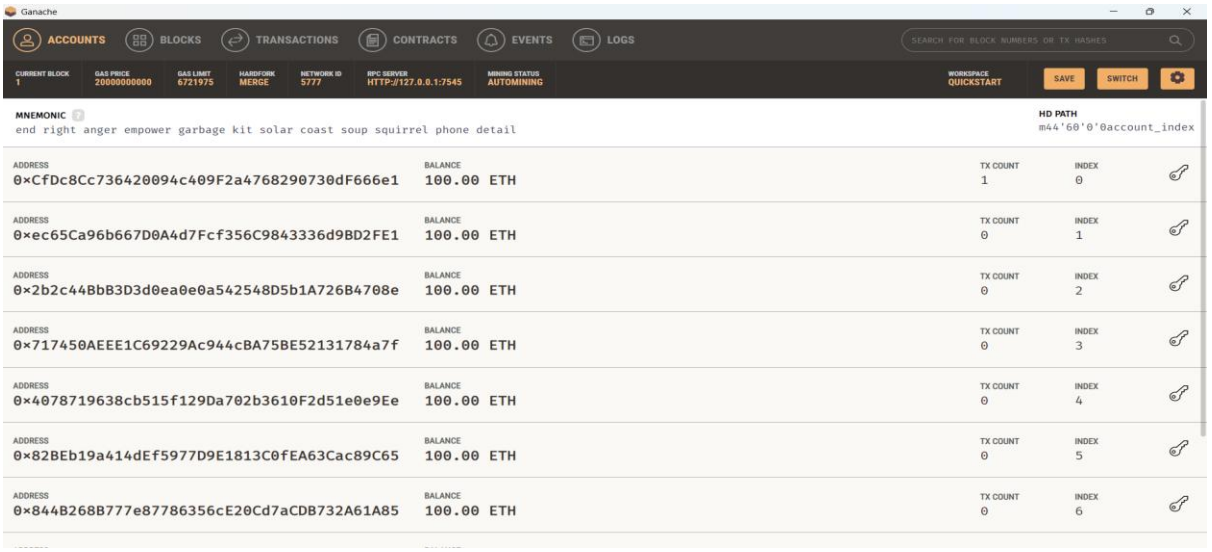

Ganache
SUPPORT GANACHE
Ganache includes Google Analytics tracking to help us better understand how you use it during your normal development practices. You can opt-out of this tracking by selecting the option below.
By enabling this feature, you provide the Truffle team with valuable metrics, allowing us to better analyze usage patterns and add new features and bug fixes faster.
Thanks for your help, and happy coding!
– The Truffle Team
WHAT WE TRACK
A unique UUID generated upon first use
Window width and height
Ganache version
Exception messages (without paths)
Screens viewed during use
We do not collect addresses or private keys.
Analytics enabled. Thanks!
CONTINUE


Ganache
v2.7.1
CREATE A WORKSPACE
Quickstart for a one-click blockchain or create a new workspace for advanced setup options.
QUICKSTART
ETHEREUM
NEW WORKSPACE
ETHEREUM


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 0
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
MNEMONIC
end right anger empower garbage kit solar coast soup squirrel phone detail
HD PATH
m44'60'0'0account_index
ADDRESS 0×CfDc8Cc736420094c409F2a4768290730dF666e1 BALANCE 100.00 ETH TX COUNT 0 INDEX 0
ADDRESS 0×ec65Ca96b667D0A4d7Fcf356C9843336d9BD2FE1 BALANCE 100.00 ETH TX COUNT 0 INDEX 1
ADDRESS 0×2b2c44BbB3D3d0ea0e0a542548D5b1A726B4708e BALANCE 100.00 ETH TX COUNT 0 INDEX 2
ADDRESS 0×717450AEEE1C69229Ac944cBA75BE52131784a7f BALANCE 100.00 ETH TX COUNT 0 INDEX 3
ADDRESS 0×4078719638cb515f129Da702b3610F2d51e0e9Ee BALANCE 100.00 ETH TX COUNT 0 INDEX 4
ADDRESS 0×82BEb19a414dEf5977D9E1813C0fEA63Cac89C65 BALANCE 100.00 ETH TX COUNT 0 INDEX 5
ADDRESS 0×844B268B777e87786356cE20Cd7aCDB732A61A85 BALANCE 100.00 ETH TX COUNT 0 INDEX 6
22°C
Mostly clear
Search
ENG
IN
21:21
29-01-2026


demo.sol
//SPDX-License-Identifier: UNLICENSED
pragma solidity >=0.7.0;
contract demo {
string public ab = "Hello world";
}
(base) PS C:\Users\mihir\Desktop\BL exp> truffle migrate
> Saving artifacts
> Total cost: 0.00069744375 ETH
Summary
> Total deployments: 1
> Final cost: 0.00069744375 ETH


1_demo_migrate.js
const demo = artifacts.require("demo");
module.exports = function (deployer) {
deployer.deploy(demo);
};
(base) PS C:\Users\mihir\Desktop\BL exp> truffle migrate
> Saving artifacts
> Total cost: 0.00069744375 ETH
Summary
> Total deployments: 1
> Final cost: 0.00069744375 ETH


Ganache
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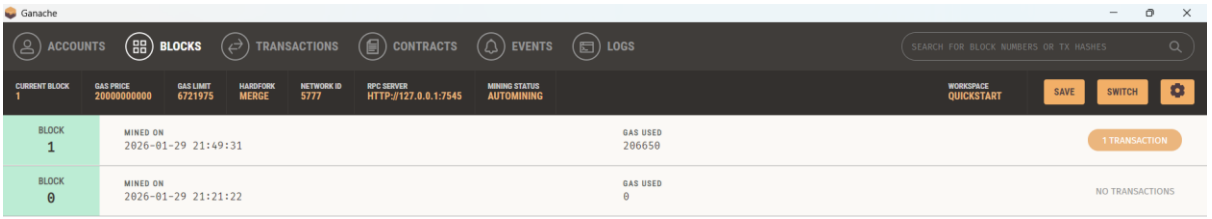
CONTRACTS
EVENTS
LOGS
MNEMONIC
end right anger empower garbage kit solar coast soup squirrel phone detail
HD PATH
m44'60'0'0account_index
0xCfDc8Cc736420094c409F2a4768290730dF666e1 100.00 ETH
0xec65Ca96b667D0A4d7Fcf356C9843336d9BD2FE1 100.00 ETH
0x2b2c44BbB3D3d0ea0e0a542548D5b1A726B4708e 100.00 ETH
0x717450AEEE1C69229Ac944cBA75BE52131784a7f 100.00 ETH
0x4078719638cb515f129Da702b3610F2d51e0e9Ee 100.00 ETH
0x82BEb19a414dEf5977D9E1813C0fEA63Cac89C65 100.00 ETH
0x844B268B777e87786356cE20Cd7aCDB732A61A85 100.00 ETH


Ganache
ACCOUNTS
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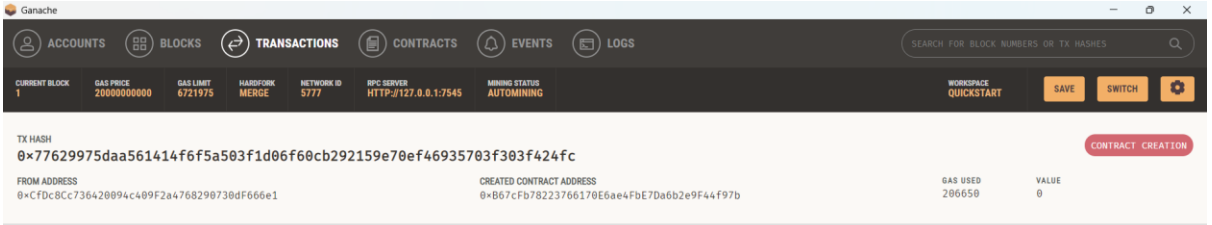
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 1
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
BLOCK 1
MINED ON 2026-01-29 21:49:31
GAS USED 206650
1 TRANSACTION
BLOCK 0
MINED ON 2026-01-29 21:21:22
GAS USED 0
NO TRANSACTIONS


Ganache
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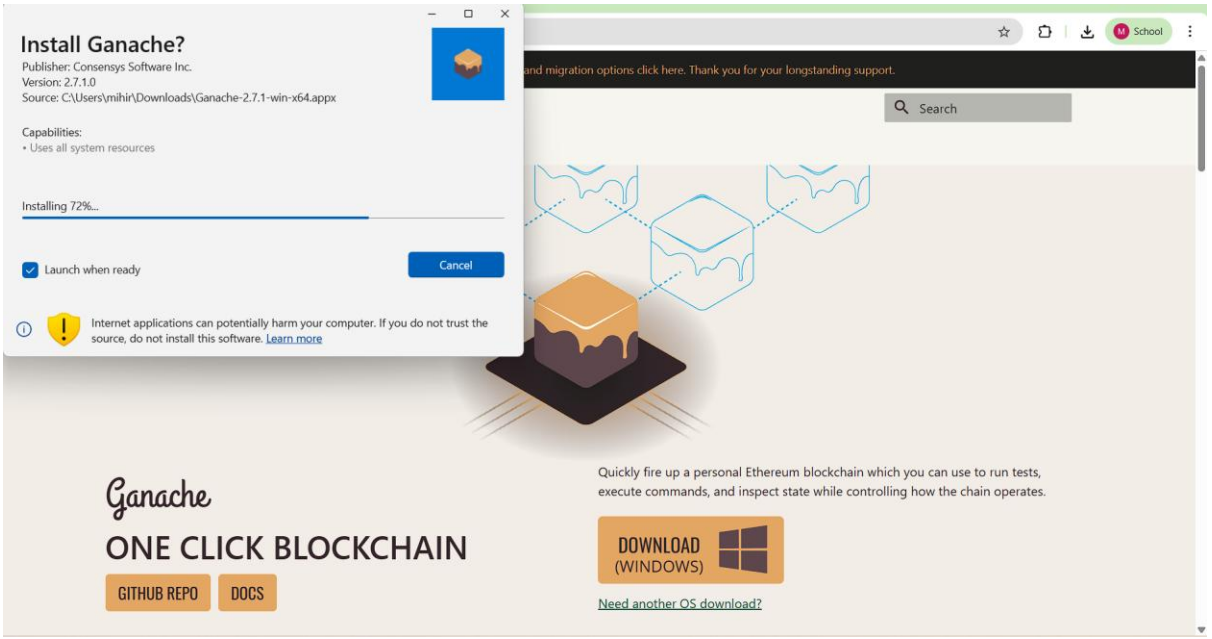
CURRENT BLOCK 1
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
TX HASH
0x77629975daa561414f6f5a503f1d06f60cb292159e70ef46935703f303f424fc
CONTRACT CREATION
FROM ADDRESS
0xCfDc8Cc736420094c409F2a4768290730dF666e1
CREATED CONTRACT ADDRESS
0xB67cFb78223766170E6ae4FbE7Da6b2e9F44f97b
GAS USED 206650
VALUE 0


Install Ganache?
Publisher: Consensys Software Inc.
Version: 2.7.1.0
Source: C:\Users\mihir\Downloads\Ganache-2.7.1-win-x64.appx
Capabilities:
• Uses all system resources
Installing 72%...
Launch when ready
Cancel
Internet applications can potentially harm your computer. If you do not trust the source, do not install this software. Learn more
School
and migration options click here. Thank you for your longstanding support.
Search
Ganache
ONE CLICK BLOCKCHAIN
GITHUB REPO
DOCS
Quickly fire up a personal Ethereum blockchain which you can use to run tests, execute commands, and inspect state while controlling how the chain operates.
DOWNLOAD (WINDOWS)
Need another OS download?

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mihir>node -v
v24.12.0

C:\Users\mihir>npm install -g truffle
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to
coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated rimraf@2.7.1: Rimraf versions prior to v4 are no longer supported
npm warn deprecated yaeti@0.0.6: Package no longer supported. Contact Support at https://www.npmjs.com/support for more info.
npm warn deprecated har-validator@5.1.5: this library is no longer supported
npm warn deprecated mkdirp-promise@5.0.1: This package is broken and no longer maintained. 'mkdirp' itself supports promises now, please switch to that.
npm warn deprecated request@2.88.2: request has been deprecated, see https://github.com/request/request/issues/3142
npm warn deprecated tar@4.4.19: Old versions of tar are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the
current version. Please update. Support for old versions may be purchased (at exhorbitant rates) by contacting i@izs.me
npm warn deprecated memdown@1.4.1: Superseded by memory-level (https://github.com/Level/community#faq)
npm warn deprecated level-errors@2.0.1: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated deferred-leveldown@5.3.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated encoding-down@6.3.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated level-js@5.0.2: Superseded by browser-level (https://github.com/Level/community#faq)
npm warn deprecated level-packager@5.1.1: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated level-codec@9.0.2: Superseded by level-transcoder (https://github.com/Level/community#faq)
npm warn deprecated levelup@4.4.0: Superseded by abstract-level (https://github.com/Level/community#faq)
npm warn deprecated glob@7.2.0: Glob versions prior to v9 are no longer supported
npm warn deprecated leveldown@5.6.0: Superseded by classic-level (https://github.com/Level/community#faq)
npm warn deprecated multibase@0.6.1: This module has been superseded by the multiformats module
npm warn deprecated multicodec@0.5.7: This module has been superseded by the multiformats module
npm warn deprecated cids@0.7.5: This module has been superseded by the multiformats module
npm warn deprecated apollo-server-errors@3.3.1: The `apollo-server-errors` package is part of Apollo Server v2 and v3, which are now end-of-life (as of Octo
ber 22nd 2023 and October 22nd 2024, respectively). This package's functionality is now found in the `@apollo/server` package. See https://www.apollographql
.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-plugin-base@3.7.2: The `apollo-server-plugin-base` package is part of Apollo Server v2 and v3, which are now end-of-life (
as of October 22nd 2023 and October 22nd 2024, respectively). This package's functionality is now found in the `@apollo/server` package. See https://www.apo
llographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-env@4.2.1: The `apollo-server-env` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22
nd 2023 and October 22nd 2024, respectively). This package's functionality is now found in the `@apollo/utils.fetcher` package. See https://www.apollographq
l.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-datasource@3.3.2: The `apollo-datasource` package is part of Apollo Server v2 and v3, which are now end-of-life (as of October 22
nd 2023 and October 22nd 2024, respectively). See https://www.apollographql.com/docs/apollo-server/previous-versions/ for more details.
npm warn deprecated apollo-server-types@3.8.0: The `apollo-server-types` package is part of Apollo Server v2 and v3, which are now end-of-life (as of Octobe
```

```
added 1174 packages in 2m

107 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New minor version of npm available! 11.0.0 -> 11.8.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.8.0
npm notice To update run: npm install -g npm@11.8.0
npm notice

C:\Users\mihir>truffle version
Truffle v5.11.5 (core: 5.11.5)
Ganache v7.9.1
Solidity v0.5.16 (solc-js)
Node v24.12.0
Web3.js v1.10.0
```

1.8 Conclusion:

The installation and configuration of Truffle provide a foundational environment for Ethereum smart contract development. By simplifying compilation, deployment, testing, and network management, Truffle enables developers to build secure and efficient decentralized applications. This experiment familiarizes students with essential blockchain development tools and prepares them for real-world DApp development.